

Outline of the Derivation of the Method of Regularized Stokeslets

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For details of the Method of Regularized Stokeslets in two dimensions, see Cortez [1]; for the three-dimensional case, see Cortez–Fauci–Medovikov [2].

1 Motivation for the Method of Regularized Stokeslets

Stokes flows play a central role in many physical and biological systems, particularly at small length scales or in highly viscous fluids. The classical Stokes equations for incompressible viscous flow are

$$-\nabla p + \mu \Delta \mathbf{u} + \mathbf{F} = \mathbf{0}, \quad \nabla \cdot \mathbf{u} = 0,$$

where μ denotes the fluid viscosity, p the pressure, and $\mathbf{u}$ the velocity. A body force $\mathbf{F}$ concentrated at a single point $\mathbf{x}_0$ is modeled by

$$\mathbf{F}(\mathbf{x}) = \mathbf{f}_0 \delta(\mathbf{x} - \mathbf{x}_0),$$

where $\mathbf{f}_0$ is a constant force vector and δ is the Dirac delta distribution.

The resulting velocity field $\mathbf{u}(\mathbf{x})$ is given by the **Stokeslet**, the fundamental solution to the Stokes equations in free space:

$$u_i(\mathbf{x}) = \frac{1}{8\pi\mu} \left(\frac{\delta_{ij}}{r} + \frac{r_i r_j}{r^3} \right) f_{0,j},$$

where $\mathbf{r} = \mathbf{x} - \mathbf{x}_0$, $r = \|\mathbf{r}\|$, and i, j index vector components.

Singularity issue

The Stokeslet velocity field becomes *singular* as $\mathbf{x} \rightarrow \mathbf{x}_0$, which presents several issues:

- The velocity grows like $1/r$, making it **numerically unstable** or undefined at the force location.
- Singular integrals are difficult to evaluate in boundary integral methods.
- In biological models (e.g., flagella and cilia), forces are often applied at many points very close to each other or to surfaces.

The Method of Regularized Stokeslets

To overcome this, the **Method of Regularized Stokeslets (MRS)** replaces the Dirac delta with a smooth, radially symmetric “blob” function:

$$\mathbf{F}(\mathbf{x}) = \mathbf{f}_0 \phi_\epsilon(\mathbf{x} - \mathbf{x}_0),$$

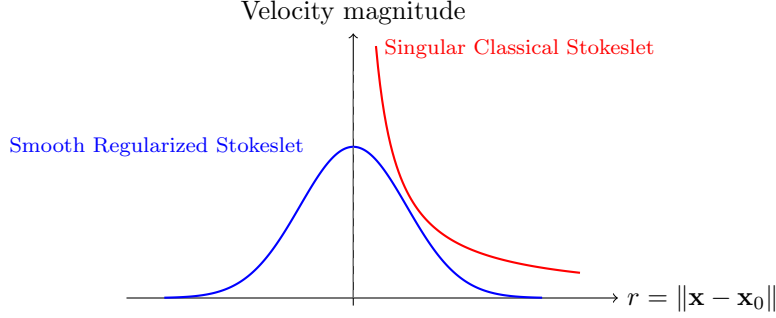
where ϕ_ϵ satisfies $\int_{\mathbb{R}^n} \phi_\epsilon(\mathbf{x}) d\mathbf{x} = 1$, and the blob size $\epsilon > 0$ controls the spread.

The velocity field can be written as

$$u_i(\mathbf{x}) = S_{ij}^\epsilon(\mathbf{x}, \mathbf{x}_0) f_{0,j},$$

where S^ϵ is the *regularized Stokeslet kernel*. This yields a **regularized Stokeslet**, a smooth approximation of the classical Stokeslet.

Illustration: Classical vs. Regularized Stokeslet



Red: Classical solution diverges at $r = 0$; Blue: Regularized solution remains smooth.

Summary

The Method of Regularized Stokeslets provides a principled and practical way to remove the singularity in the classical solution of the Stokes equations. It is especially crucial for numerical simulations in biological fluid dynamics where point forces are used extensively.

2 Solutions of Regularized Steady Stokes Equations

The method of regularized Stokeslets is based on exact solutions to the steady Stokes equations with body forces modeled as smooth, localized functions that respect the incompressibility condition. These equations, in either two or three dimensions, are:

$$\mu \Delta \mathbf{u} = \nabla p - \mathbf{F}(\mathbf{x}), \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (2)$$

where μ is viscosity, p is pressure, $\mathbf{u}$ is velocity, and $\mathbf{F}$ is the body force (force density).

We model $\mathbf{F}$ as a smoothed version of a point force $\mathbf{f}_0$ distributed over a small region centered at $\mathbf{x}_0$:

$$\mathbf{F}(\mathbf{x}) = \mathbf{f}_0 \phi_\epsilon(\mathbf{x} - \mathbf{x}_0), \quad (3)$$

where ϕ_ϵ is a radially symmetric smoothing function (“blob”) satisfying $\int \phi_\epsilon(\mathbf{x}) d\mathbf{x} = 1$.

Typical examples include:

- In $\mathbb{R}^2$:

$$\phi_\epsilon(\mathbf{x}) = \frac{1}{\pi \epsilon^2} e^{-|\mathbf{x}|^2/\epsilon^2}, \quad \phi_\epsilon(\mathbf{x}) = \frac{3\epsilon^3}{2\pi(|\mathbf{x}|^2 + \epsilon^2)^{5/2}}.$$

- In $\mathbb{R}^3$:

$$\phi_\epsilon(\mathbf{x}) = \frac{3}{4\pi\epsilon^3} e^{-|\mathbf{x}|^2/\epsilon^2}, \quad \phi_\epsilon(\mathbf{x}) = \frac{15\epsilon^4}{8\pi(|\mathbf{x}|^2 + \epsilon^2)^{7/2}}.$$

A typical blob is shown in Figure 1 for two values of ϵ . As ϵ decreases, the blob becomes narrower and taller to preserve the total integral of 1. In the limit $\epsilon \rightarrow 0$, ϕ_ϵ converges to the Dirac delta function.

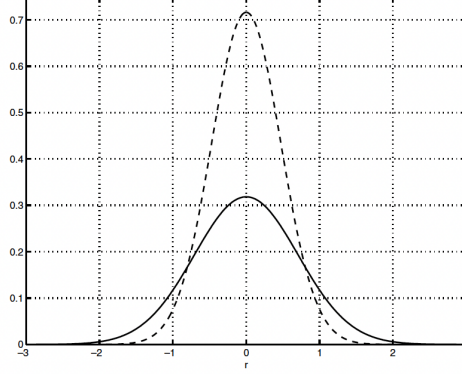


Figure 1: Typical blob for two different values of ϵ . Source: Cortez [1]

To solve for p and $\mathbf{u}$ in Equations (1)–(2) with force distribution (3), we define:

- $G_\epsilon(\mathbf{x})$: solution of $\Delta G_\epsilon = \phi_\epsilon(\mathbf{x})$ in $\mathbb{R}^n$,
- $B_\epsilon(\mathbf{x})$: solution of $\Delta B_\epsilon = G_\epsilon(\mathbf{x})$ in $\mathbb{R}^n$.

a) Derivation of Pressure

Taking divergence of Equation (1) gives:

$$\begin{aligned}\nabla \cdot (\mu \Delta \mathbf{u}) &= \nabla \cdot (\nabla p - \mathbf{F}) \\ \Rightarrow \mu \Delta (\nabla \cdot \mathbf{u}) &= \Delta p - \nabla \cdot \mathbf{F} \\ \Rightarrow \Delta p &= \nabla \cdot \mathbf{F} = \mathbf{f}_0 \cdot \nabla \phi_\epsilon.\end{aligned}$$

Using $\Delta G_\epsilon = \phi_\epsilon$ implies $\nabla \phi_\epsilon = \nabla(\Delta G_\epsilon)$, hence

$$\Delta p = \mathbf{f}_0 \cdot \nabla(\Delta G_\epsilon) \Rightarrow p = \mathbf{f}_0 \cdot \nabla G_\epsilon,$$

which determines the pressure up to an additive constant.

b) Derivation of Velocity

From above, $\nabla p = (\mathbf{f}_0 \cdot \nabla) \nabla G_\epsilon$. Substituting into Equation (1) yields:

$$\mu \Delta \mathbf{u} = (\mathbf{f}_0 \cdot \nabla) \nabla G_\epsilon - \mathbf{f}_0 \phi_\epsilon.$$

Since $\Delta B_\epsilon = G_\epsilon$, we have $\nabla G_\epsilon = \nabla(\Delta B_\epsilon)$, and therefore

$$\begin{aligned}\mu \Delta \mathbf{u} &= (\mathbf{f}_0 \cdot \nabla) \nabla(\Delta B_\epsilon) - \mathbf{f}_0 \Delta G_\epsilon \\ \Rightarrow \mu \mathbf{u} &= (\mathbf{f}_0 \cdot \nabla) \nabla B_\epsilon - \mathbf{f}_0 G_\epsilon.\end{aligned}$$

which defines the velocity due to a regularized Stokeslet (up to a constant vector field).

c) Multiple Forces

For N point forces $\mathbf{f}_k$ located at $\mathbf{x}_k$, the pressure and velocity fields are:

$$p(\mathbf{x}) = \sum_{k=1}^N \mathbf{f}_k \cdot \nabla G_\epsilon(\mathbf{x} - \mathbf{x}_k) \quad (6)$$

$$\mathbf{u}(\mathbf{x}) = \mathbf{U}_0 + \frac{1}{\mu} \sum_{k=1}^N [(\mathbf{f}_k \cdot \nabla) \nabla B_\epsilon(\mathbf{x} - \mathbf{x}_k) - \mathbf{f}_k G_\epsilon(\mathbf{x} - \mathbf{x}_k)] \quad (7)$$

where the constant flow $\mathbf{U}_0$ will be determined later.

Let $r = \|\mathbf{x}\|$. Using radial symmetry:

$$\nabla G_\epsilon = G'_\epsilon(r) \frac{\mathbf{x}}{r}, \quad \nabla B_\epsilon = B'_\epsilon(r) \frac{\mathbf{x}}{r},$$

yields

$$(\mathbf{f} \cdot \nabla) \nabla B_\epsilon = \mathbf{f} \frac{B'_\epsilon(r)}{r} + (\mathbf{f} \cdot \mathbf{x}) \mathbf{x} \frac{r B''_\epsilon(r) - B'_\epsilon(r)}{r^3}.$$

Let $r_k = \|\mathbf{x} - \mathbf{x}_k\|$ and rewrite Equations (6)–(7) as:

$$p(\mathbf{x}) = \sum_{k=1}^N [\mathbf{f}_k \cdot (\mathbf{x} - \mathbf{x}_k)] \frac{G'_\epsilon(r_k)}{r_k} \quad (8)$$

$$\mathbf{u}(\mathbf{x}) = \mathbf{U}_0 + \frac{1}{\mu} \sum_{k=1}^N \left\{ \mathbf{f}_k \left[\frac{B'_\epsilon(r_k)}{r_k} - G_\epsilon(r_k) \right] + [\mathbf{f}_k \cdot (\mathbf{x} - \mathbf{x}_k)] (\mathbf{x} - \mathbf{x}_k) \left[\frac{r_k B''_\epsilon(r_k) - B'_\epsilon(r_k)}{r_k^3} \right] \right\} \quad (9)$$

In two dimensions, one can verify that

$$\frac{B'_\epsilon(r)}{r} - G_\epsilon(r) = -\frac{1}{2} G_\epsilon(r) - \frac{1}{8\pi},$$

which implies that a residual constant flow remains. To eliminate this constant flow, one can choose

$$\mathbf{U}_0 = \frac{1}{8\pi\mu} \sum_k \mathbf{f}_k.$$

In three dimensions, the regularized formulation does not introduce a constant offset, so one may simply take $\mathbf{U}_0 = \mathbf{0}$.

3 Explicit 2D Regularized Stokeslet Solutions for a Given Blob Function

In $\mathbb{R}^2$, choose the blob:

$$\phi_\epsilon(\mathbf{x}) = \frac{3\epsilon^3}{2\pi(|\mathbf{x}|^2 + \epsilon^2)^{5/2}}.$$

a) Derivation of G_ϵ

First we use radial symmetry and the Laplacian in polar coordinates:

$$\Delta G_\epsilon = \frac{1}{r} [r G'_\epsilon(r)]' = \phi_\epsilon(r).$$

then integrate

$$[r G'_\epsilon(r)]' = r \phi_\epsilon(r) \Rightarrow r G'_\epsilon(r) = \int_0^r s \phi_\epsilon(s) ds = I_1 \Rightarrow G'_\epsilon(r) = \frac{I_1}{r} = I_2 \Rightarrow G_\epsilon(r) = \int I_2 dr.$$

This yields

$$G_\epsilon(r) = \frac{1}{2\pi} \left[\ln(\sqrt{r^2 + \epsilon^2} + \epsilon) - \frac{\epsilon}{\sqrt{r^2 + \epsilon^2}} \right].$$

b) Derivation of B'_ϵ

Since $\Delta B_\epsilon = G_\epsilon(r)$, integrate:

$$[rB'_\epsilon(r)]' = rG_\epsilon(r) \Rightarrow rB'_\epsilon(r) = \int_0^r sG_\epsilon(s) ds = I_3 \Rightarrow B'_\epsilon(r) = \frac{I_3}{r}$$

which gives

$$B'_\epsilon(r) = \frac{1}{8\pi} \left[2r \ln \left(\sqrt{r^2 + \epsilon^2} + \epsilon \right) - r - \frac{2r\epsilon}{\sqrt{r^2 + \epsilon^2} + \epsilon} \right].$$

c) Final Expressions for Pressure and Velocity

Substituting into Equations (8) and (9) yields

$$p(\mathbf{x}) = \sum_{k=1}^N \frac{1}{2\pi} [\mathbf{f}_k \cdot (\mathbf{x} - \mathbf{x}_k)] \left[\frac{r_k^2 + 2\epsilon^2 + \epsilon\sqrt{r_k^2 + \epsilon^2}}{(\sqrt{r_k^2 + \epsilon^2} + \epsilon)(r_k^2 + \epsilon^2)^{3/2}} \right] \quad (10)$$

$$\begin{aligned} \mathbf{u}(\mathbf{x}) = & \sum_{k=1}^N -\frac{\mathbf{f}_k}{4\pi\mu} \left[\ln \left(\sqrt{r_k^2 + \epsilon^2} + \epsilon \right) - \frac{\epsilon(\sqrt{r_k^2 + \epsilon^2} + 2\epsilon)}{(\sqrt{r_k^2 + \epsilon^2} + \epsilon)\sqrt{r_k^2 + \epsilon^2}} \right] \\ & + \frac{1}{4\pi\mu} [\mathbf{f}_k \cdot (\mathbf{x} - \mathbf{x}_k)] (\mathbf{x} - \mathbf{x}_k) \left[\frac{\sqrt{r_k^2 + \epsilon^2} + 2\epsilon}{(\sqrt{r_k^2 + \epsilon^2} + \epsilon)^2 \sqrt{r_k^2 + \epsilon^2}} \right] \end{aligned} \quad (11)$$

where $r_k = \|\mathbf{x} - \mathbf{x}_k\|$.

As $\epsilon \rightarrow 0$, the regularized Stokeslets converge to the classical (singular) Stokeslet solutions:

$$p(\mathbf{x}) = \sum_{k=1}^N \frac{[\mathbf{f}_k \cdot (\mathbf{x} - \mathbf{x}_k)]}{2\pi r_k^2} \quad (12)$$

$$\mathbf{u}(\mathbf{x}) = \sum_{k=1}^N \left[-\frac{\mathbf{f}_k}{4\pi\mu} \ln(r_k) + \frac{[\mathbf{f}_k \cdot (\mathbf{x} - \mathbf{x}_k)](\mathbf{x} - \mathbf{x}_k)}{4\pi\mu r_k^2} \right] \quad (13)$$

which are singular at the locations of the point forces.

d) Linear Matrix System in 2D

Equation (11) can be written as a linear system

$$\mathbf{U} = M\mathbf{F}, \quad (14)$$

where $\mathbf{U} \in \mathbb{R}^{2N_x}$ and $\mathbf{F} \in \mathbb{R}^{2N_y}$ are stacked velocity and force vectors, and $M \in \mathbb{R}^{2N_x \times 2N_y}$ is the regularized Stokeslet matrix. Here, $\mathbf{U}$ contains the velocities evaluated at target points $\mathbf{x}_i$, $i = 1, \dots, N_x$, and $\mathbf{F}$ contains the point forces applied at source points $\mathbf{y}_j$, $j = 1, \dots, N_y$.

Vector assembly. Let the target points be $\{\mathbf{x}_i\}_{i=1}^{N_x} \subset \mathbb{R}^2$ and the source points be $\{\mathbf{y}_j\}_{j=1}^{N_y} \subset \mathbb{R}^2$. Stack the velocity components and forces as

$$\mathbf{U} = \begin{bmatrix} u_1(\mathbf{x}_1) \\ u_2(\mathbf{x}_1) \\ \vdots \\ u_1(\mathbf{x}_{N_x}) \\ u_2(\mathbf{x}_{N_x}) \end{bmatrix} \in \mathbb{R}^{2N_x}, \quad \mathbf{F} = \begin{bmatrix} f_1(\mathbf{y}_1) \\ f_2(\mathbf{y}_1) \\ \vdots \\ f_1(\mathbf{y}_{N_y}) \\ f_2(\mathbf{y}_{N_y}) \end{bmatrix} \in \mathbb{R}^{2N_y}.$$

Pairwise geometry and kernels. For each pair (i, j) , set

$$\mathbf{r}_{ij} = \mathbf{x}_i - \mathbf{y}_j = (\Delta x_{ij}, \Delta y_{ij})^\top, \quad r_{ij} = \|\mathbf{r}_{ij}\|, \quad A_{ij} = \sqrt{r_{ij}^2 + \varepsilon^2}.$$

Define the scalar functions

$$H(r_{ij}) = \ln(A_{ij} + \varepsilon) - \frac{\varepsilon(A_{ij} + 2\varepsilon)}{(A_{ij} + \varepsilon)A_{ij}}, \quad L(r_{ij}) = \frac{A_{ij} + 2\varepsilon}{(A_{ij} + \varepsilon)^2 A_{ij}},$$

.

Block matrix structure. The 2×2 block mapping the force at $\mathbf{y}_j$ to the velocity at $\mathbf{x}_i$ is

$$M_{ij} = \frac{1}{4\pi\mu} \begin{bmatrix} -H(r_{ij}) + L(r_{ij}) \Delta x_{ij}^2 & L(r_{ij}) \Delta x_{ij} \Delta y_{ij} \\ L(r_{ij}) \Delta x_{ij} \Delta y_{ij} & -H(r_{ij}) + L(r_{ij}) \Delta y_{ij}^2 \end{bmatrix}.$$

Assemble the full matrix $M \in \mathbb{R}^{2N_x \times 2N_y}$ in 2×2 blocks:

$$M = \begin{bmatrix} M_{11} & M_{12} & \cdots & M_{1N_y} \\ M_{21} & M_{22} & \cdots & M_{2N_y} \\ \vdots & \vdots & \ddots & \vdots \\ M_{N_x 1} & M_{N_x 2} & \cdots & M_{N_x N_y} \end{bmatrix}, \quad \mathbf{U} = M \mathbf{F}.$$

In $\mathbb{R}^3$, the derivation proceeds analogously; see Cortez–Fauci–Medovikov [2] for a standard choice of blob and the resulting expressions for G_ε , B_ε , pressure, and velocity.

4 References

- [1] R. Cortez, “The Method of Regularized Stokeslets,” SIAM Journal on Scientific Computing, 23(4) (2001): 1204–1225
- [2] R. Cortez, L. Fauci, and A. Medovikov, “The Method of Regularized Stokeslets in Three Dimensions: Analysis, Validation, and Application to Helical Swimming,” Physics of Fluids, 17 (2005): 1–14